

omicstra

project data: TNBC-92

Cohort: Wang et al. 2024 TNBC, 94 patients, 260 matched subarrays, ~278k niches. **Unit:** niche (k=6 spatial neighbors), ~1200 cells. **Evaluation:** subarray-level, patient-stratified 85/15 split. **FDR control:** BH-FDR over the test families described per hypothesis.

method definitions

Method	Definition
R4	InfoNCE cross-attention alignment. ST queries attend over H&E tile tokens (center + 6 spatial neighbors). produces a weighted-sum H&E representation conditioned on ST.
R1	InfoNCE late-fusion alignment with independent modality projections (mean-pooled H&E + MLP, novae + gpath2vec + MLP). standard CLIP-style contrastive baseline.
R6	R1 ablation: drops gpath2vec embeddings from the ST input. tests whether H&E pathway signal depends on the pathway anchor being present on the ST side.
R2	Supervised contrastive late-fusion alignment using soft mc_weights similarity as targets (instead of binary match/mismatch).
R3	Barlow Twins late-fusion: redundancy-reduction loss that pushes cross-correlation matrix toward identity. no explicit negatives.
B1	Canonical Correlation Analysis. classical, linear, closed-form. maximizes cross-modal correlation in whitened space.
B2	Orthogonal Procrustes. classical, linear. finds the orthogonal rotation that best aligns the two modalities' PCA-projected spaces.
B3	Unaligned PCA. independent per-modality PCAs with L2-norm. no cross-modal objective. sanity floor.
B4	Random-initialized untrained MLPs. R1 architecture with Xavier init, no training. strict baseline that isolates the contribution of contrastive training from the architecture alone.

metric definitions

Metric	What it measures
R@K (recall at K)	for each query niche, rank candidates by cosine similarity. fraction of queries where the true match appears in the top K. used for H1 retrieval evaluation.
MRR (mean reciprocal rank)	average of 1/rank across queries. rewards both high ranks and consistency. used alongside R@K.
AUC	probability that a random matched pair has higher cosine similarity than a random mismatched pair. 0.5 = chance, 1.0 = perfect. scale-invariant; primary H1 metric.
alignment gap	mean(matched cosines) minus mean(mismatched cosines). positive = matched pairs systematically closer. unbiased; no chance-floor convention issue.
CKA (Centered Kernel Alignment)	structural alignment between modalities: do the H&E and ST manifolds have similar pairwise-similarity structure overall? complements pointwise alignment.
ARI (Adjusted Rand Index)	cluster-vs-label agreement, corrected for chance. ARI=1 perfect, ARI=0 random. used for H2 structural coherence on Wang's mc_megacluster labels.

silhouette score	compactness of clusters vs separation between clusters. high silhouette = well-separated, compact clusters.
LP accuracy (linear probe)	train a logistic regression to predict biological label from the embedding on a patient-stratified split. tests whether biology is linearly separable in the embedding, independent of cluster geometry.
matched-pair cosine	within a tissue compartment, average cosine similarity between paired H&E and ST niches. tests whether biology is consistent across modalities within compartments.
bio z / patient z	permutation z-scores: how much more extreme is observed cross-modal coherence with respect to biological label (or patient ID) than under a shuffled null? ratio = bio z / patient z measures biology vs leakage.
CCA correlation (canonical correlation analysis)	for a single pathway target, find the linear projection of the embedding most correlated with the pathway score. tests whether a specific biological signal lives along an interpretable latent direction.
percent significant cells	at DAG depth, fraction of (run x view x pathway-node) cells that pass BH-FDR after cross-patient sub-split CCA + permutation null. measures generalizable fine-grained pathway encoding.

proposal alignment and results

each row maps one hypothesis or objective from the original proposal to: what we evaluated, what we observed, what it means, the metrics used, the methods that performed best, and the baselines that anchored the comparison.

H1 . Cross-Modal Alignment

Field	Detail
Proposal alignment	Directly matched the proposal goal of aligning H&E and spatial transcriptomics embeddings within a shared latent space using contrastive learning.
Evaluation question	Can H&E embeddings retrieve matched ST niches more accurately than unaligned or classical alignment methods?
Observed result	R4 achieved strongest alignment with AUC 0.851 and CKA 0.564. R1 and R6 also exceeded the proposal target range with AUC 0.733 to 0.741. classical baselines spanned 0.443 to 0.703; B4 untrained sat at chance (0.491).
Interpretation	Cross-attention alignment contributed genuine cross-modal biological alignment beyond manifold compression. a rank-matched CKA control (R1 projected to top-3 PCs reaches CKA 0.234 vs R4 full 0.564) rejects the "R4 wins by collapsing the manifold" explanation.
Scientific conclusion	Contrastive cross-attention alignment captured stronger cross-modal biological structure than classical baselines. the contrastive training is doing the bulk of the work (B4 at chance); cross-attention adds the rest.
Primary metrics	Recall@K, MRR, AUC, CKA, alignment gap
Strong methods	R4 (best), R1, R6
Baselines	B1, B2, B3, B4

H1 . Structural Agreement

Field	Detail
Proposal alignment	Matched the proposal objective of evaluating latent geometry agreement and representation structure across modalities.
Evaluation question	Do aligned embeddings exhibit stronger cross-modal structural agreement than baseline methods?
Observed result	Contrastive runs separated cleanly from classical baselines on CKA: range 0.120 to 0.564 vs 0.086 to 0.124, no overlap. R4 gap +0.195 vs B3 gap -0.049 (sanity floor anti-correlation).
Interpretation	Matched-versus-mismatched separation increased substantially after contrastive alignment, and the gain in CKA reflects genuine manifold structure agreement, not just pointwise matching.
Scientific conclusion	Contrastive objectives improved latent geometry alignment beyond linear correlation methods.
Primary metrics	CKA, alignment gap, cosine separation
Strong methods	R4
Baselines	B1, B3

H2 . Structural Coherence

Field	Detail
Proposal alignment	Directly evaluated the proposal hypothesis that aligned latent spaces preserve biologically meaningful tumor microenvironment organization.
Evaluation question	Do the proposal's nine spatial archetypes cluster more coherently in the aligned latent space?
Observed result	The proposal's archetype labels turned out to be patient-level pseudobulk: all niches in a patient inherit one archetype. ARI on archetype is a patient classification test in disguise, and classical baselines that amplify patient identity (B1 ARI 0.307) score highest. on the proper niche-level test (Wang's mc_megacluster, 14 classes, per-spot NMF), R1 and R6 achieve KMeans MC ARI 0.231 to 0.251. R4 scores low on KMeans (0.098) but passes the patient-stratified linear probe at parity (0.230 vs R1 0.260, raw H&E 0.239).
Interpretation	R4 still encodes biology - a linear classifier trained on R4's embedding predicts biological labels just as well as one trained on R1's embedding or on raw H&E features. but R4's manifold is tight enough that distance-based clustering (KMeans) can't separate the biology into clean groups. R1 and R6 produce wider embeddings where biologically similar niches form visibly separate clusters.
Scientific conclusion	R1 and R6 are the right choice when you need to group niches into biologically distinct regions on a tissue section (immune-rich, stromal, necrotic, TLS, etc.). R4 trades that cluster geometry for tighter manifold structure, which is what makes it generalize across patients on H3. fusion-axis trade-off rather than a single winner.
Primary metrics	ARI, silhouette score, LP accuracy, KNN purity
Strong methods	R1, R6 (KMeans + LP), R4 (LP only)
Baselines	B1, B3 (lose 31 to 37 percent of ARI on the resolution shift, quantifying the patient-leakage component)

H2 . Compartment Contrasts

Field	Detail
Proposal alignment	Matched the proposal goal of compartment-level analysis across tumor microenvironment regions.
Evaluation question	Do morphologically distinct compartments (tumor, TLS, necrosis) show higher matched-pair cosine similarity than ambiguous stromal compartments (high-TIL, low-TIL)?
Observed result	Only the TLS vs TIL-stroma contrasts consistently matched the directional prediction (passes in R1, R2, R6 with $z=+3.2$ to $+5.3$). Tumor and necrosis contrasts inverted decisively ($z \leq -2.8$).
Interpretation	The large Tumor ($n=4384$) and Necrosis ($n=1178$) annotations pool subtypes and dilute matched-pair similarity; the small well-curated TIL annotations ($n=105$ to 354) have higher within-compartment homogeneity and inflate cosine. this is an annotation-vocabulary issue, not a biology-absence issue.
Scientific conclusion	Compartment-level alignment quality depended strongly on annotation granularity and morphological purity. an 18-class compartment vocabulary on the 90-subarray annotated subset is queued.
Primary metrics	Matched-pair cosine similarity by compartment, Welch z-test
Strong methods	R1, R6 (on the TLS contrasts)
Baselines	B1 (all six contrasts non-significant)

H2 . Biological vs Patient Signal

Field	Detail
Proposal alignment	Extended the proposal by explicitly evaluating biological signal preservation relative to patient leakage.
Evaluation question	Does the aligned latent space encode biology more strongly than patient identity?
Observed result	R4 has the lowest patient z (22.97) in the grid, the strongest patient suppression on z_he. B1 amplifies patient identity 3.3x (patient z 122.13 vs raw 37.25), making it actively anti-helpful for cohort-level interpretation.
Interpretation	Linear probes recovered biology even when clustering metrics underestimated manifold structure. R4's strong patient suppression is what makes it transfer well across patients (see H3).
Scientific conclusion	Shared embeddings preserved biological organization while suppressing patient-specific signal. B1 is the only run that amplifies patient identity more than biology and should be avoided for any interpretation task that requires cross-patient validity.
Primary metrics	Patient-ID decoding, bio z / patient z ratio
Strong methods	R4 (patient suppression), R6 (bio/patient ratio 0.548, highest)
Baselines	B1 (anti-helpful, ratio 0.064 below raw-modality floor 0.129)

H3 . Pathway Interpretability

Field	Detail
Proposal alignment	Directly matched the proposal objective of evaluating pathway-associated biological signal within the shared latent geometry.
Evaluation question	Do Reactome pathway embeddings (TGF-beta, Immune System, ECM Organization, Cell Cycle, Programmed Cell Death) correlate with directions in the shared latent space?
Observed result	At the 5-parent test fit on all test niches, all 120 cells (8 runs x 3 views x 5 pathways) pass BH-FDR < 0.001. universally supported. when restricted to cross-patient sub-split, R4 achieves 15 of 15 significant transfers while R1, R2, R3 achieve only 5 of 15.
Interpretation	The proposal-literal test is supported universally but does not discriminate. the cross-patient sub-split is the discriminating test; R4 dominant, classical baselines strong (12 to 14 of 15), late-fusion contrastive runs fail.
Scientific conclusion	Shared latent geometry preserved transferable pathway-associated biological signal. R4's tight manifold encoding generalizes across held-out patients; R1's wider manifold encodes patient-specific gradients that do not transfer.
Primary metrics	Univariate canonical correlation per pathway, permutation null, BH-FDR
Strong methods	R4 (cross-patient winner), B1/B2/B3 (strong cross-patient), R1/R2/R3 (fit-on-all only)
Baselines	B1 fails the related Bareche bio-vs-patient test; B2 and B3 pass the cross-patient CCA

H3 . Cross-Patient Transfer (DAG decomposition)

Field	Detail
Proposal alignment	Matched the proposal goal of evaluating generalization across held-out patients, at sub-pathway resolution. extends the 5-parent test (Reactome top-level subtrees) down to the full DAG of pathway children to see whether transferable signal lives at coarse or fine granularity.
Evaluation question	Does pathway-associated latent structure transfer across patients at fine-grained sub-pathway resolution? we extended the test from 5 top-level parents to 395 DAG nodes under the 5 named parents (gene-set size 3 to 500).
Observed result	Across 9480 cells (8 runs x 3 views x 395 nodes): R4 leads with 9.9 percent significant; classical baselines competitive (B2 9.3 percent, B1 8.8 percent, B3 8.0 percent); R6 4.6 percent; R1 3.4 percent; R2 2.9 percent; R3 2.3 percent. R4 wins or ties on every named parent (Cell Cycle 69 percent, ECM 93 percent, Immune 83 percent, PCD 75 percent, TGF-beta 81 percent).
R4 top sub-pathway hits	Co-inhibition by BTLA (immune checkpoint, $z=6.92$); Interleukin receptor SHC signaling ($z=6.80$); Integrin cell surface interactions ($z=6.21$); ECM proteoglycans ($z=5.73$); Regulation of IFNA/IFNB signaling ($z=5.71$); Elastic fibre formation ($z=5.33$); Collagen biosynthesis ($z=5.28$); Collagen degradation ($z=5.22$); FLT3 signaling via SRC family kinases ($z=5.15$); Calcineurin activates NFAT ($z=5.10$). these are biologically distinct sub-programs spanning immune checkpoint biology and stromal/collagen remodeling.
Interpretation	The 5-parent specificity off-diagonal of 0.819 (which would suggest aliased single-axis encoding) is a coarse-aggregation artifact: parent unions of overlapping Reactome subtrees share gene content and produce correlated AUCell scores. at DAG resolution R4 encodes distinct clinically interpretable sub-programs.
Scientific conclusion	Cross-attention alignment improved pathway-level biological transferability and the transferable encoding spans biologically distinct sub-pathways relevant to TNBC: immune checkpoint expression (clinically relevant to checkpoint-inhibitor response) and stromal/collagen remodeling (clinically relevant to TME architecture and metastasis).
Primary metrics	Held-out cross-patient CCA, permutation null + BH-FDR over 9480 cells, percent significant per run and per parent
Strong methods	R4 (wins overall and at every parent)
Baselines	B1, B2, B3 competitive at fine resolution; R1, R6 near floor despite winning H2 niche-level

Systems-Oriented Evaluation

Field	Detail
Proposal alignment	Strongly aligned with the proposal framing of multimodal alignment as an orchestration and evaluation problem rather than a single-model optimization problem.
Evaluation question	Did one alignment strategy dominate every biological objective simultaneously?
Observed result	No single method dominated every hypothesis family. R4 wins H1 cross-modal retrieval, H3 cross-patient pathway transfer, and H3 sub-pathway resolution. R1 and R6 win at grouping niches into biologically distinct regions on a tissue section (H2). classical baselines are competitive at fine pathway detection but fail H1 retrieval and amplify patient identity on H3-extended (B1 specifically).
Interpretation	Different alignment architectures optimize different biological objectives. niche-level cluster geometry, cross-modal retrieval, and cross-patient pathway transfer are not co-monotonic.
Scientific conclusion	The framework exposed biologically meaningful trade-offs supporting task-conditional routing rather than a single globally optimal embedding strategy.
Primary metrics	Integrated evaluation across H1, H2, H3 families
Strong methods	R4 (retrieval, cross-patient transfer); R1, R6 (region clustering on tissue)
Baselines	all baselines lose at least one hypothesis family decisively

task-conditional routing rule

the matrix collapses to this routing rule for downstream use:

If the question is	Route to	Evidence
cross-modal retrieval (find matched ST niche from H&E)	R4	H1 AUC 0.851, CKA 0.564, rank-matched control passes
group niches that share similar tumor-microenvironment state (immune-hot, stromal, necrotic, TLS, etc.) to find biologically meaningful regions in a tissue section	R1 or R6	H2 KMeans on mc_megacluster (Wang 14-class niche-level NMF): R6 ARI 0.251, R1 ARI 0.231; R1 linear probe accuracy 0.260
amplify biology, suppress patient identity	R6 or R1; R4 for leakage minimization	bio/patient ratio: R6 0.548, R1 0.517; R4 patient z 22.97 lowest
find pathway-similar niches across new patients	R4	H3 cross-patient 15/15; H3 DAG 9.9% leading
interpret fine-grained sub-pathway biology	R4 (with B1/B2 competitive on ECM and Immune)	H3 DAG: R4 wins or ties every parent; B1/B2 reach 98 percent on ECM
avoid for cohort-level biology interpretation	B1 (CCA)	H3-extended bio/patient ratio 0.064 below raw-modality floor 0.129

no single method dominates every objective; the framework's value is in exposing the trade-offs so that downstream analyses route the right question to the right method. this is what the omicstra alignment-agent will encode as task-conditional dispatch. when this evaluation runs on additional cohorts (HEST, future TNBC studies, other tumor types) and each run's final matrix is persisted back to the alignment-agent's rules store, a routing pattern accumulates: which question types map reliably to which method families, where the trade-offs hold, and where they shift with cohort characteristics. the value of omicstra as an MCP server compounds with each cohort - the agent gets sharper at routing because it carries forward what previous evaluations learned. TNBC-92 is the first row.

consolidated proposal-alignment matrix

Hypothesis / Objective	Proposal Alignment	Evaluation Question	Observed Result	Interpretation	Scientific Conclusion	Primary Metrics	Strong Methods	Baselines
H1 Cross-Modal Alignment	Directly matched the proposal goal of aligning H&E and ST embeddings within a shared latent space using contrastive learning.	Can H&E embeddings retrieve matched ST niches more accurately than unaligned or classical methods?	R4 strongest with AUC 0.851, CKA 0.564. R1 and R6 in target range (AUC 0.733 to 0.741). B4 untrained at chance (0.491).	Cross-attention contributed genuine cross-modal biological alignment. rank-matched CKA control (R1 at rank 3 = 0.234 vs R4 full 0.564) rejects manifold-collapse.	Contrastive cross-attention captured stronger cross-modal biological structure than classical baselines.	R@K, MRR, AUC, CKA, alignment gap	R4, R1, R6	B1, B2, B3, B4
H1 Structural Agreement	Matched the proposal objective of evaluating latent geometry agreement across modalities.	Do aligned embeddings exhibit stronger cross-modal structural agreement than baselines?	Contrastive runs separated from classical baselines on CKA: 0.120 to 0.564 vs 0.086 to 0.124, no overlap. R4 gap +0.195 vs B3 -0.049.	Matched-vs-mismatched separation increased substantially after contrastive alignment; CKA gain reflects genuine manifold agreement.	Contrastive objectives improved latent geometry alignment beyond linear correlation methods.	CKA, alignment gap, cosine separation	R4	B1, B3
H2 Structural Coherence	Directly evaluated the proposal hypothesis that aligned latent spaces preserve biologically meaningful TME organization.	Do the proposal's nine spatial archetypes cluster more coherently in the aligned space?	Archetype labels are patient-level pseudobulk - ARI on archetype is a patient classification test. On niche-level mc_megacluster: R6 ARI 0.251, R1 0.231; R4 KMeans 0.098 but LP at parity (0.230).	R4 encodes biology (linear probe at parity with R1 and raw H&E) but its tight manifold prevents KMeans from separating biology into clean groups. R1/R6 give visibly separable cluster regions.	R1 and R6 are right for grouping niches into biologically distinct regions on a tissue section. R4 trades cluster geometry for tighter manifold structure that enables cross-patient transfer (H3).	ARI, silhouette, LP accuracy, kNN purity	R1, R6 (KMeans+LP); R4 (LP only)	B1, B3
H2 Compartment Contrasts	Matched the proposal goal of compartment-level analysis across TME regions.	Do morphologically distinct compartments (tumor, TLS, necrosis) show higher matched-pair cosine than ambiguous stroma (high-TIL, low-TIL)?	Only TLS vs TIL-stroma contrasts matched the directional prediction (z=+3.2 to +5.3 in R1/R2/R6). Tumor and necrosis contrasts inverted (z ≤ -2.8).	Large Tumor (n=4384) and Necrosis (n=1178) annotations pool subtypes and dilute matched-pair similarity; small well-curated TIL annotations (n=105 to 354) have higher within-compartment homogeneity.	Compartment-level alignment quality depended on annotation granularity and morphological purity. 18-class compartment vocabulary on the 90-subarray annotated subset is queued.	Matched-pair cosine, Welch z-test	R1, R6 (TLS contrasts)	B1
H2 Biology vs Patient	Extended the proposal by explicitly evaluating biological signal preservation vs patient leakage.	Does the aligned space encode biology more strongly than patient identity?	R4 has lowest patient z (22.97) - strongest patient suppression on z_he. B1 amplifies patient identity 3.3x (patient z 122.13 vs raw 37.25).	Linear probes recovered biology even where clustering underestimated structure. R4's patient suppression is what enables H3 cross-patient transfer.	Shared embeddings preserved biological organization while suppressing patient-specific signal. B1 amplifies patient identity more than biology and should be avoided for cohort-level interpretation.	Patient-ID decoding, bio z / patient z ratio	R6 (ratio 0.548), R1 (0.517), R4 (patient suppression)	B1 (anti-helpful, ratio 0.064 below raw 0.129)
H3 Pathway Interpretability	Directly matched the proposal objective of evaluating pathway-associated biological signal within shared latent geometry.	Do Reactome pathways (TGF-beta, Immune, ECM, Cell Cycle, PCD) correlate with shared-latent directions?	Fit-on-all: 120/120 cells pass BH-FDR (universally supported). Cross-patient sub-split: R4 15/15 significant; R1/R2/R3 only 5/15; classical 12-14/15.	The fit-on-all test is supported universally but does not discriminate. The cross-patient sub-split is the discriminating test; R4 dominant, late-fusion contrastive runs fail.	Shared latent geometry preserved transferable pathway signal. R4's tight manifold generalizes across patients; R1's wider manifold encodes patient-specific gradients that don't transfer.	Univariate CCA per pathway, permutation null, BH-FDR	R4 (cross-patient); B1/B2/B3 (12-14/15 cross-patient)	R1/R2/R3 fail cross-patient (5/15)
H3 Cross-Patient Transfer (DAG)	Matched the proposal goal of evaluating generalization across held-out patients, at sub-pathway resolution.	Does pathway latent structure transfer at sub-pathway resolution? extends 5-parent test to 395 DAG nodes.	9480 cells (8 runs x 3 views x 395 nodes): R4 leads at 9.9% significant; B2 9.3%, B1 8.8%, B3 8.0%; R6 4.6%, R1 3.4%. R4 wins/ties every parent (Cell Cycle 69%, ECM 93%, Immune 83%, PCD 75%, TGF-beta 81%).	R4 top hits are biologically distinct sub-programs: BTLA checkpoint (z=6.92), IL receptor SHC (6.80), Integrin (6.21), IFN-alpha/beta (5.71), Collagen biosynthesis/degradation (5.22-5.28), NFAT (5.10). 5-parent off-diagonal 0.819 is a gene-set-overlap artifact, not encoding property.	Cross-attention improved pathway-level transferability across biologically distinct sub-programs spanning immune-checkpoint biology (relevant to checkpoint-inhibitor response) and stromal/collagen remodeling (relevant to TME architecture).	Held-out cross-patient CCA + permutation BH-FDR over 9480 cells	R4 (wins overall and at every parent)	B1, B2, B3 competitive at fine resolution; R1/R6 near floor
Systems-Oriented Evaluation	Strongly aligned with the proposal framing of multimodal alignment as orchestration and evaluation, not single-model optimization.	Did one alignment strategy dominate every biological objective simultaneously?	No. R4 wins H1 retrieval, H3 cross-patient, H3 sub-pathway. R1/R6 win H2 tissue-region clustering. Classical baselines competitive at fine pathway detection but fail H1 retrieval; B1 amplifies patient identity.	Different alignment architectures optimize different biological objectives. niche-level cluster geometry, cross-modal retrieval, and cross-patient pathway transfer are not co-monotonic.	The framework exposed biologically meaningful trade-offs supporting task-conditional routing rather than a single globally optimal embedding strategy.	Integrated H1, H2, H3	R4, R1, R6 (each wins different families)	All baselines lose at least one family decisively